

BACHELOR OF SCIENCE IN COMPUTER SCIENCE AND ENGINEERING

Orthographic Gaussian Splatting for Volumetric Anatomical Visualization

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Declaration of Candidate

This is to certify that the work presented in this thesis is the outcome of the analysis and experiments carried out by **Tasreef Mahmood Ridun**, **MD Mushfiqur Rahman Mushfique**, and **Kazi Jawadul Islam Jishan** under the supervision of **Tareque Mohmud Chowdhury**, Assistant Professor, Department of Computer Science and Engineering, Islamic University of Technology (IUT), OIC, Islamic University of Technology, Dhaka, Bangladesh. It is also declared that neither this thesis nor any part of it has been submitted anywhere else for any degree or diploma. Information derived from the published and unpublished work of others have been acknowledged in the text and a list of references is given.

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Dedicated to to the people of Palestine

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Acknowledgement

Acknowledgments

We are profoundly grateful to our supervisor, Tareque Mohmud Chowdhury, for his endless patience, insightful critique, and His expertise and encouragement were instrumental in the completion of this thesis.

Finally, to our parents, whose love and support have been a constant source of strength. Thank you for always being there.

To all of you, we extend our heartfelt appreciation.

Abstract

Computed tomography (CT) produces stacks of cross-sectional images that form volumetric data, but clinicians usually view them as separate slices or simple 3D models lacking interactivity. Current 3D rendering methods assume perspective cameras and repeated image features, which CT data does not provide. This thesis presents an orthographic Gaussian splatting pipeline: we generate orthographic camera parameters, skip feature matching, convert CT slices into Gaussian splats, and optimize their arrangement to reconstruct coherent volumes. The work includes data conversion and parameter adjustment. Tests on clinical head CT scans and a synthetic dataset show interactive renderings with accurate anatomical shapes and density patterns at low computational cost. This approach can speed up CT review and surgical planning.

Chapter 1

Introduction

The field of medical imaging stands at the confluence of technology and life sciences, offering unprecedented insights into the internal structures of the human body. Among various modalities, Computed Tomography (CT) has established itself as a crucial tool, producing sequential cross-sectional images that together form detailed volumetric datasets. Yet, despite advances in imaging hardware, the visualization techniques used to interpret these volumes have remained largely static — often confined to manual slice-by-slice exploration or basic three-dimensional reconstructions with significant limitations in fidelity and interactivity.

In parallel, the domain of computer graphics and machine learning has witnessed a revolution in 3D scene representation through methods like Neural Radiance Fields (NeRF) and, more recently, 3D Gaussian Splatting. These approaches offer highly efficient and photorealistic rendering by representing scenes as distributions of learnable primitives, enabling real-time novel view synthesis without explicit surface modeling. However, these methods are typically optimized for natural, feature-rich environments and rely fundamentally on perspective projections and feature continuity across views — assumptions that are starkly violated in clinical CT data.

This research addresses the critical methodological gap by proposing a novel adaptation of Gaussian Splatting tailored for volumetric medical visualization under orthographic projection conditions. Unlike traditional scene reconstruction pipelines, the proposed framework synthesizes orthographic camera models and bypasses the need for feature-matching-based Structure-from-Motion. Through careful preprocessing, pipeline engineering, and optimization adjustments, the adapted system achieves coherent three-dimensional reconstructions from sequential CT slices, preserving anatomical structures with fidelity while operating under significantly constrained visual in-

formation.

The significance of this work lies in its potential to reshape how medical volumetric data is visualized and interpreted. By enabling lightweight, real-time volumetric reconstructions that require no external ground truth models or complex manual interventions, this method can improve diagnostic workflows, enhance anatomical understanding, and lay the groundwork for new classes of interactive medical applications.

In summary, this thesis presents an innovative orthographic Gaussian Splatting framework for medical volumes, bridging contemporary advances in 3D graphics with critical unmet needs in medical imaging visualization. The outcomes demonstrate both the technical feasibility and the promising clinical relevance of the proposed approach.

1.1 What Sections that are Included in this chapter

The Introduction chapter serves as the backbone of the thesis, contextualizing the motivation behind the research, articulating the central problem, identifying the key challenges, and highlighting the specific contributions made. It aims to guide the reader through a structured understanding of why this research was necessary, what challenges were addressed, and how the research systematically approached the problem.

The sections included in this chapter are:

- **Introductory Information**, to establish the context and background of the study.
- **Motivations and Scope**, to clarify the driving factors and boundaries of the research.
- **Problem Statement**, to define the core research problem with precision.
- **Research Challenges**, to identify technical and conceptual hurdles faced.
- **Contribution**, to enumerate the original inputs of this research into the academic discourse.
- **Organization**, to outline the structure of the remaining chapters.

Each section is crafted to maintain a clear, logical progression that aligns with academic expectations for a graduate thesis.

1.2 Introductory Information

Medical imaging technologies have revolutionized the landscape of modern healthcare, enabling physicians and researchers to visualize internal human anatomy non-invasively with extraordinary precision. Techniques such as Computed Tomography (CT) provide detailed volumetric data by capturing sequential 2D slices across different axes. Despite this wealth of information, the standard modes of interacting with such data — typically through slice-by-slice exploration or basic volume renderings — remain limited in terms of interactivity, depth perception, and spatial comprehension.

Simultaneously, the field of computer graphics and machine learning has witnessed significant advancements in 3D scene reconstruction. Techniques like Neural Radiance Fields (NeRF) and its derivatives have demonstrated impressive capabilities for photorealistic novel view synthesis. Among these, 3D Gaussian Splatting has emerged as a computationally efficient, real-time method for rendering dynamic 3D scenes using learned Gaussian primitives rather than dense volumetric grids or meshes.

However, Gaussian Splatting pipelines are typically designed under the assumption of perspective projection and feature-rich environments — conditions inherently absent in raw medical datasets. This misalignment presents a substantial challenge when attempting to directly apply these pipelines to clinical data. In particular, CT datasets, characterized by orthographic projections and limited inter-slice visual continuity, violate the core assumptions upon which Gaussian Splatting optimization strategies are built.

The research presented in this thesis focuses on bridging this methodological gap. Specifically, it aims to adapt the Gaussian Splatting framework to the unique requirements of volumetric medical visualization, leveraging orthographic projections to reconstruct coherent 3D anatomical structures from CT slice datasets.

1.3 Motivations and Scope

The motivation for this research arises at the intersection of two critical observations: technological limitations within current visualization pipelines for medical imaging, and the growing demand in healthcare for intuitive, high-fidelity, volumetric recon-

structions.

From a technological standpoint, while existing Gaussian Splatting techniques have shown remarkable success in reconstructing outdoor scenes, indoor environments, and even synthetic 3D objects, they fundamentally rely on certain assumptions that are incompatible with clinical datasets. Traditional SfM (Structure from Motion) tools presume that multiple images capture the same object from different viewpoints, identifying common features (edges, textures, patterns) across views to solve camera poses. In contrast, sequential CT slices depict entirely different cross-sections of anatomy with minimal shared features, rendering standard SfM inapplicable.

From a clinical standpoint, there is a pressing need to improve how volumetric data is visualized. Physicians are often forced to mentally reconstruct three-dimensional structures from two-dimensional slices, a process that is both cognitively demanding and error-prone. An accessible, real-time volumetric visualization could greatly enhance diagnostic accuracy, improve preoperative planning, and deepen anatomical understanding.

1.4 Scope of Research

Given these motivations, this research focuses on:

- Designing preprocessing pipelines to convert raw CT slice data into formats compatible with splatting frameworks.
- Synthesizing camera intrinsics and extrinsics based on orthographic projections, bypassing the need for traditional feature-based SfM.
- Modifying and fine-tuning the Gaussian Splatting optimization process to better handle the characteristics of medical datasets.
- Exploring volumetric visualization strategies that remain computationally lightweight and clinically viable.

While the focus remains on CT datasets, the methodology and findings are expected to lay groundwork adaptable to other slice-based imaging modalities, such as MRI or PET scans.

1.5 Problem Statement

The core problem addressed by this research stems from the fundamental incompatibility between the assumptions underlying existing Gaussian Splatting pipelines and the nature of medical CT datasets.

Specifically, existing pipelines presuppose:

- The availability of perspective-projected views from various angles around a scene.
- The presence of consistent visual features across views to facilitate accurate camera pose estimation through Structure-from-Motion.
- An optimization process that refines 3D point distributions based on perspective cues and photometric consistency.

In contrast, CT datasets:

- Are composed of sequential slices representing orthographic projections rather than perspective images.
- Exhibit negligible feature continuity between adjacent slices, making feature-based camera solving infeasible.
- Do not inherently provide camera parameters, requiring external estimation or simulation.

Thus, the central research problem can be articulated as:

"How can the principles of 3D Gaussian Splatting be adapted to enable volumetric reconstructions from orthographically projected, feature-sparse medical slice datasets, thereby achieving coherent and high-fidelity medical visualizations?"

This main problem encompasses several sub-problems:

- How to generate synthetic, yet physically meaningful, camera models for CT slices.
- How to initialize and optimize Gaussian splats in the absence of perspective cues.
- How to validate the quality of the volumetric reconstruction in clinically relevant terms.

1.6 Research Challenges

Throughout the course of this research, numerous technical and conceptual challenges were encountered and addressed. The principal challenges include:

Dataset Preparation and Conversion

CT datasets, especially high-resolution clinical datasets, are typically stored in specialized formats such as JPEG2000 or DICOM. These formats, while optimized for medical archiving and transmission, are not directly compatible with most computer vision pipelines. Conversion into usable formats (TIF, PNG) was necessary, with careful attention to preserving spatial resolution and bit-depth fidelity.

Moreover, slice alignment and ordering had to be meticulously verified, as errors at this stage would propagate significant inaccuracies in downstream volumetric reconstructions.

Camera Parameter Synthesis

Standard Structure-from-Motion pipelines utterly failed on CT datasets due to the absence of shared visual features between slices. Consequently, camera intrinsics and extrinsics had to be synthetically generated, assuming orthographic projections and modeling inter-slice spacing based on known or estimated slice thicknesses.

An initial mistake — assuming perspective projection — led to chaotic and incoherent reconstructions, emphasizing the critical importance of correctly modeling the imaging geometry.

Pipeline Adaptation and Debugging

The base Gaussian Splatting pipeline obtained from public repositories was incomplete, poorly documented, and incompatible with modern system libraries. Significant engineering effort was invested into:

- Resolving dependency mismatches (CUDA, PyTorch versions).
- Correcting deprecated code functionalities.
- Removing non-essential dependencies, such as background image requirements.
- Rewiring the pipeline to accept non-standard data inputs.

This iterative debugging process formed a significant portion of the research workload, highlighting the often underappreciated engineering challenges in experimental machine learning research.

Optimization Behavior under Orthographic Assumptions

Gaussian splatting optimization relies heavily on learning scene geometry based on multi-view parallax under perspective projection. Under orthographic assumptions, this parallax is absent, leading to poor convergence behaviors:

- Central structures collapsed or disappeared.
- Peripheral structures dominated the optimization.
- Initialization randomness propagated through training.

Addressing these issues required experimenting with extended training iterations, modified density thresholds, and exploration of density voxel initialization strategies to provide better starting conditions for the optimizer.

Visualization and Interpretation

Standard 3D visualization libraries (e.g., MeshLab, Open3D) were insufficient for displaying Gaussian splats effectively. Web-based tools had to be adopted, and custom scripts were developed for basic point cloud visualization and slicing, allowing for preliminary qualitative assessments.

1.7 Contribution

This thesis presents several novel contributions that advance the state-of-the-art in applying Gaussian Splatting techniques to medical imaging:

- **Orthographic Camera Synthesis Framework**

Developed a flexible and parametric method for generating camera extrinsics and intrinsics tailored to orthographic slice datasets, bypassing the need for Structure-from-Motion.

- **Adaptation of Cinematic Gaussian Splatting Pipeline**

Successfully modified and adapted an incomplete cinematic splatting pipeline to accept orthographic inputs and handle non-standard data types encountered in clinical datasets.

- **Proof-of-Concept Volumetric Reconstructions**

Demonstrated preliminary volumetric reconstructions from both synthetic datasets (e.g., M60 Tank dataset) and converted CT slice stacks, validating the feasibility of the proposed adaptations.

- **Integration of Density Voxel Preprocessing**

Integrated external density voxel generation (via Napier) to assist in better Gaussian initialization, addressing optimization collapse issues under orthographic projections.

- **Identification of Future Pathways**

Laid the groundwork for future improvements, including DICOM-specific processing, GPU-based rasterization for real-time rendering, and potential fine-tuning of splatting techniques for clinical deployment.

Chapter 2

Gaussian Splat

A Gaussian splat is a technique used in 3D rendering and visualization, where 3D data points are represented not as discrete points or voxels, but as blurred or smooth "splats" that have a Gaussian distribution. Essentially, it's a method of approximating a surface or volume by projecting Gaussian functions into 2D space, creating a soft, circular blur-like effect for each data point. These splats can represent either the shape or color of a surface, contributing to the overall appearance when viewed from different perspectives.

2.1 What is a Gaussian Splat?

Main Goal

The primary objective is to enable **real-time, high-quality rendering** (e.g., 30+ FPS at 1080p) of 3D scenes from multi-view captures, without requiring extensive training times typically associated with NeRF models.

Approach

The method involves the following innovations:

- A **novel scene representation**: A cloud of **anisotropic 3D Gaussians**.
- A **differentiable rasterizer**: A fast, visibility-aware, tile-based rendering engine optimized for GPU performance.

This approach combines the rendering efficiency of GPU-optimized game engines with the realism typically associated with NeRF models.

Part 1: 3D Gaussian Representation

Instead of using voxels, MLPs, or dense point clouds, the authors utilize:

- **3D Gaussians**, where each Gaussian is characterized by:
 - A 3D **position**.
 - A **covariance matrix** (which controls its shape, akin to a flexible ellipsoid).
 - **Opacity** (alpha).
 - **Spherical Harmonics (SH) coefficients** for color, which aid in view-dependent lighting.

Anisotropic Gaussians are more versatile than simple spherical blobs — they stretch and compress to model geometry such as thin rails or leaves.

Notably, this approach eliminates the need for normals, bypassing a typically complex geometric step.

Part 2: Optimization & Density Control

Given the collection of Gaussians, optimization is required. This process can be likened to sculpting a statue from raw material:

- **Stochastic Gradient Descent (SGD):**
 - Optimizes each Gaussian’s position, shape, color, and opacity.
 - The loss function combines **L1** error and **D-SSIM** to maintain sharpness and structural fidelity.
- **Adaptive Densification:**
 - Areas of the scene that are **under-reconstructed** are enhanced by cloning Gaussians.
 - Areas that are **over-reconstructed** are refined by splitting large Gaussians into smaller ones.

- Gaussians with minimal contribution (e.g., transparent or overly large Gaussians) are **pruned**.
 - This is a dynamic optimization process — Gaussians are created, adjusted, and removed to best fit the scene.
-

Part 3: Fast Differentiable Tile Rasterizer

The rendering process is optimized through the following steps:

- The screen is divided into **16×16 tiles**.
- For each tile, the Gaussians that **project into it** are gathered and sorted by depth.
- A **GPU radix sort** ensures that proper visibility (front-to-back) is maintained.
- **Alpha-blending** is applied, similar to the method used for semi-transparent objects in real-time graphics.

The key advantages of this approach include:

- It is **differentiable**, allowing it to work seamlessly with backpropagation during training.
 - There is no upper limit on the number of Gaussians that can influence a given pixel, making it suitable for complex scenes.
-

Part 4: Results

When compared to **Mip-NeRF360**, **InstantNGP**, and **Plenoxels**, the system demonstrates:

- Comparable or superior **visual quality**.
 - Rendering speeds of up to **135 FPS**.
 - Training times reduced to **minutes**, as opposed to hours.
-

Part 5: Limitations

Despite its successes, the system has certain limitations:

- **Artifacts** in areas that were poorly captured or are not visible in the dataset.
 - **Spotty or elongated Gaussians** in certain situations.
 - **Popping** effects when large Gaussians shift visibility order.
 - **High memory usage** during training, though this can be optimized.
-

In Summary

Gaussian Splatting presents a hybrid approach:

- Combining **explicit** techniques (such as point clouds, which provide speed).
- With the **smoothness and differentiability** of NeRFs (which excel at learning from images).

This represents one of the closest approximations to real-time, high-fidelity 3D graphics generated from photographs — without the need for manual modeling or extensive pre-processing.

2.2 3D Gaussian Representation

This section provides a detailed, rigorous explanation of the components that define a 3D Gaussian. Each component plays a critical role in how these Gaussians cooperate to reconstruct a scene from multi-view images.

1. Mean – The Position in 3D Space

The **mean** represents the simplest and most fundamental aspect of a 3D Gaussian. It specifies the **location** of the Gaussian in 3D space.

One can conceptualize the mean as the **center point** of a soft, flexible shape. For example, if reconstructing a chair, the Gaussian may be placed near one of its legs. Each Gaussian is positioned at a specific (x, y, z) coordinate in the scene.

2. Covariance Matrix – The Shape and Orientation

The **covariance matrix** defines the **shape**, **size**, and **orientation** of the Gaussian.

While the mean locates the Gaussian, the covariance matrix governs the **geometry** of the Gaussian. It controls:

- Whether the Gaussian is spherical, elliptical, or cigar-shaped.
- The tilt or rotation of the Gaussian.

A **3×3 covariance matrix** in 3D controls the following aspects:

- The extent of the Gaussian along each of the three axes (x, y, z).
- The degree to which the Gaussian is rotated or skewed.

This characteristic makes the Gaussians **anisotropic**, meaning their shape can vary in different directions, providing flexibility to model complex geometries accurately:

- A thin object, like a table leg, is represented by a stretched-out Gaussian.
- A flat surface, such as a wall, is represented by a wide, flat Gaussian.
- A small corner detail is represented by a tight, compact Gaussian.

Since the covariance matrix is differentiable, it can be **optimized** smoothly during training.

3. Opacity – How Much Light It Lets Through

The **opacity** of a Gaussian determines how **transparent or opaque** it is.

- If $\alpha = 1$, the Gaussian is **fully opaque**, meaning it contributes entirely to the pixel it covers.
- If $\alpha = 0$, the Gaussian is **invisible** and does not contribute to the rendered image.
- Values between 0 and 1 allow for smooth blending between overlapping Gaussians.

This concept can be compared to layering sheets of colored plastic: the more opaque the sheets, the more they dominate the visible scene, while the more transparent they are, the more they blend with the layers beneath.

The opacity parameter enables control over the visibility and density of Gaussians, ensuring that:

- Gaussians representing transparent materials, such as air, are more transparent.
 - Gaussians near solid objects, like walls, are more opaque.
-

4. Color via Spherical Harmonics (SH) – Lighting That Changes with View

The color of each Gaussian varies depending on the viewer’s angle. This is modeled using **Spherical Harmonics (SH)**, a mathematical framework that allows the representation of how the color changes with the viewing direction.

- Spherical Harmonics store a compact set of color values, which encode how the Gaussian’s color changes as a function of the viewing angle.
- This approach is smooth, differentiable, and ideal for real-time rendering.

This representation enables Gaussians to exhibit **view-dependent appearance**, which is essential for realistic rendering. For instance, a shiny surface will reflect light differently when viewed from the front compared to a steep angle.

How These Four Components Work Together

Each 3D Gaussian can be visualized as a **small, intelligent light-reflecting volume**. The key components of each Gaussian are summarized in the following table:

Component	Function	Visual Analogy
Mean (μ)	Determines the position of the Gaussian in space	The center point of the volume
Covariance (Σ)	Defines the shape and orientation of the Gaussian	The geometry of the volume (e.g., spherical, elongated)
Opacity (α)	Controls the transparency or opacity of the Gaussian	The degree of transparency of the volume
SH (color)	Adjusts the color based on the viewing angle	How the Gaussian reflects light from different perspectives

When combined, **millions of these Gaussians** allow for the reconstruction of an entire 3D scene, not through surfaces but by using **soft, colorful, view-aware volumes**.

Color via Spherical Harmonics (SH) – Lighting That Varies with View

The Problem: A Single Color is Insufficient

Consider the task of recreating a shiny metal teapot in 3D. Suppose you capture photographs of the teapot from different angles: front, back, top, and sides. Upon inspection, you observe the following:

- From the **front**, it appears bright.
- From the **side**, reflections make it appear darker.
- From **above**, it may reflect the ceiling.

In summary, **the perceived color varies depending on the viewer's position**. This phenomenon, known as **view-dependent appearance**, plays a significant role in making a 3D scene appear realistic.

Now, if one were to assign a constant color (e.g., $\text{RGB} = [0.5, 0.5, 0.5]$) to every point in the scene, the color would remain fixed irrespective of the viewing angle. This approach results in a scene that is **flat**, **unnatural**, and **unrealistic**.

The Solution: Spherical Harmonics

Rather than assigning a fixed color to each Gaussian, the method utilizes **Spherical Harmonics (SH)** to represent how color changes depending on the viewing direction.

What Are Spherical Harmonics (Intuitively)?

Spherical Harmonics can be viewed as:

- A **mathematical tool** used to describe functions over a sphere (i.e., directions in 3D space).
- Analogous to Fourier series, which represent functions over time, Spherical Harmonics describe functions over angles.

Spherical Harmonics enable us to express the following:

"In this direction, the color is slightly brighter. In that direction, it appears darker."

The significant advantage of Spherical Harmonics is that only a **small set of coefficients** is required to capture complex directional behaviors.

How Do They Work in Gaussian Splatting?

In the context of Gaussian Splatting, each 3D Gaussian stores a set of SH coefficients that encode color as a function of viewing direction.

The process works as follows:

1. To render a pixel, a ray is cast from the camera.
2. The Gaussians that intersect with this ray are identified.
3. For each intersecting Gaussian:
 - The direction from the center of the Gaussian to the camera is computed (i.e., the viewpoint).
 - The SH function is evaluated for that direction.
 - A color value is obtained, which is tailored to the specific viewpoint.
4. The colors from all overlapping Gaussians are blended to produce the final pixel color.

In effect, each Gaussian provides information about its appearance from various perspectives:

"From your current position, I appear slightly more blue and less red!"

Why Not Use Neural Networks for This?

Some earlier methods, such as NeRF, utilize neural networks (MLPs) to predict view-dependent color. However, this approach has some limitations:

- MLPs are typically **slow**, operate as **black-box models**, and are not always stable.
- Debugging MLPs can be challenging.
- They are not optimized for fast GPU-based rendering.

In contrast, Spherical Harmonics offer several advantages:

- They are **simple**, **compact**, and **efficient to evaluate**.

- They perform exceptionally well on GPUs.
- They are **differentiable**, making them suitable for gradient-based training.

How Many SH Coefficients Are Used?

Typically, four bands of SH are used, which correspond to the following:

- Band 1: 1 coefficient (representing the base or diffuse color).
- Band 2: 4 coefficients.
- Band 3: 9 coefficients.
- Band 4: 16 coefficients per color channel.

Thus, for each Gaussian, 48 coefficients are used (16 for each of the R, G, and B channels). Despite the large number of coefficients, this system is capable of running in real time.

How Is SH Used During Training?

At the beginning of the training process, only the lowest SH band is optimized, providing a rough estimate of the base color. As training progresses, higher bands are gradually incorporated, allowing for the capture of more nuanced view-dependent effects. This **progressive refinement** contributes to the stability of the training process.

Visual Example

Imagine standing in front of a shiny red sports car:

- From the **front**, the hood reflects a bright red hue.
- From the **side**, the chrome trim reflects the sky.
- From the **back**, the tail lights emit a distinct glow.

A single RGB value is insufficient to capture all of these variations. However, an SH function can describe the color variations from all directions. In this way, each Gaussian becomes a dynamic light-reflector, responding differently based on the observer's position.

What are Spherical Harmonics Mathematically?

Spherical Harmonics are a set of orthonormal basis functions defined on the surface of a sphere. These functions serve as a generalization of sine and cosine functions, but rather than representing time, they describe functions over directions (angles on a sphere).

Any function defined on the unit sphere, such as how color or light varies with direction, can be approximated using Spherical Harmonics.

Spherical Harmonic Basis Functions

The basis functions are denoted as:

$$Y_l^m(\theta, \phi)$$

where:

- l is the degree, a non-negative integer ($l = 0, 1, 2, \dots$),
- m is the order, an integer within the range $-l \leq m \leq l$,
- θ is the inclination (angle from the "north pole," akin to latitude),
- ϕ is the azimuth (angle around the equator, similar to longitude).

These functions form a complete, orthonormal basis over the sphere, meaning that any function $f(\theta, \phi)$ defined on the sphere can be written as:

$$f(\theta, \phi) = \sum_{l=0}^L \sum_{m=-l}^l c_l^m Y_l^m(\theta, \phi)$$

where c_l^m are the coefficients that weight each SH basis function to shape the final result.

How Many Coefficients Are There?

The number of SH basis functions (and corresponding coefficients) for a given band L is:

$$\text{Total SH functions} = (L + 1)^2$$

Thus:

- **Band 0:** 1 function (the constant term),
- **Band 1:** 1 function (the diffuse color),
- **Band 2:** 4 functions,
- **Band 3:** 9 functions,
- **Band 4:** 16 functions.

In Gaussian Splatting, typically either Band 3 ($L = 3$) or Band 4 ($L = 4$) is used, which results in:

- 9 or 16 coefficients per color channel,
- 27 or 48 total SH coefficients per Gaussian (for RGB).

How Is SH Evaluated in Practice?

To determine the color of a Gaussian from a specific direction (e.g., toward the camera), the following steps are performed:

1. Compute the unit direction vector $\mathbf{v}$ from the Gaussian to the camera.
2. Convert $\mathbf{v}$ to spherical coordinates (θ, ϕ) .
3. Evaluate all SH basis functions $Y_l^m(\theta, \phi)$.
4. Multiply each basis function by the corresponding coefficient c_l^m .
5. Sum the results to obtain the final RGB color.

This process is repeated separately for the R, G, and B channels.

Real-Time Optimization: Use Precomputed SH Functions

To optimize rendering time, SH basis functions can be precomputed and stored as closed-form expressions, eliminating the need for recalculation during each frame. Below are examples of low-order SH basis functions (for $L = 2$):

$$Y_0^0 = \frac{1}{2}\sqrt{\frac{1}{\pi}}, \quad Y_1^{-1} = \sqrt{\frac{3}{4\pi}}y, \quad Y_1^0 = \sqrt{\frac{3}{4\pi}}z, \quad Y_1^1 = \sqrt{\frac{3}{4\pi}}x$$

Full Pipeline: From 2D Images to Trained 3D Gaussian Splatting

0. Dataset & Assumptions

We begin with the following assumptions:

- A **static 3D scene** captured via multiple 2D photographs.
- **Known camera intrinsics and extrinsics** (focal length, image size, pose).
- All images are in RGB format (some pipelines may use masks, but these are not required here).
- The scene may be either bounded (e.g., a room) or unbounded (e.g., an outdoor space).

If the camera poses are not already available:

- A **Structure-from-Motion (SfM)** algorithm (such as COLMAP) can be run to compute these poses and generate a **sparse 3D point cloud**.

It is crucial to assume that we have **calibrated cameras and sparse 3D points** as input.

1. Structure-from-Motion (SfM)

Purpose: Estimate camera intrinsics, poses, and generate a sparse 3D point cloud from the 2D images.

Inputs:

- A set of 2D RGB images.

Outputs:

- Intrinsic matrix K : focal length, principal point.
- Extrinsic matrix $[R|t]$: rotation and translation per image.
- Sparse 3D points $\{\mathbf{p}_i\}$: reconstructed scene landmarks.

This point cloud serves as the seed for the **initial 3D Gaussians**.

2. Gaussian Initialization

Each 3D point $\mathbf{p}_i \in \mathbb{R}^3$ is converted into a **3D Gaussian**.

Each Gaussian is initialized with the following:

Property	Initial Value	Description
Position (μ)	SfM point $\mathbf{p}_i$	Center in 3D space
Covariance (Σ)	Isotropic: $\sigma^2 I$	A small sphere
Opacity (α)	Small value (e.g., 0.1)	Will be optimized
SH Coefficients	Zero or small random values	Base color (view-dependent)

The covariance is stored indirectly as a scale vector $\mathbf{s}$ and rotation quaternion $\mathbf{q}$:

$$\Sigma = R \cdot S \cdot S^\top \cdot R^\top$$

where:

- R is the 3x3 rotation matrix derived from quaternion $\mathbf{q}$,
- S is the diagonal scale matrix derived from vector $\mathbf{s}$.

The total number of Gaussians is approximately equal to the number of SfM points (typically in the range of 10k–50k).

3. Training Pipeline Initialization

Preprocessing:

- Downscale the training images to **quarter resolution** for warm-up.
- Store:
 - Camera matrices $P = K \cdot [R|t]$
 - Image tensors on the GPU
- Set initial hyperparameters:
 - Learning rate per parameter type (position, opacity, SH, etc.)
 - Use SGD optimizer (e.g., Adam)

SH Setup:

- Start with **band 0** (just diffuse color).
- Gradually unlock higher bands (134) every 1000 iterations.

4. Forward Pass – Differentiable Rendering

Simulate what the camera would see when observing the Gaussians. This forms the **core renderer**, which is fully differentiable.

Step-by-step Rendering:

1. **Select a batch of camera views** (e.g., 1–4):

- This provides a batch of camera matrices P_j and corresponding RGB images I_j .

2. **Project each 3D Gaussian to 2D screen space:**

$$\mathbf{x}_i = \pi(P_j \cdot \mu_i)$$

3. **Project its covariance from 3D to 2D** (as an ellipse):

$$\Sigma' = JW\Sigma W^\top J^\top$$

4. **Cull and assign Gaussians to tiles:**

- Divide the screen into 16x16 tiles.
- For each Gaussian, determine which tiles it touches via bounding ellipse checks.

5. **Sort Gaussians per tile by depth:**

- Compute view-space depth of each Gaussian center.
- Sort using GPU radix sort.

6. **Rasterize per-tile using alpha blending:**

$$C_{\text{pixel}} = \sum_i T_i \cdot \alpha_i \cdot c_i \quad \text{where} \quad T_i = \prod_{j=1}^{i-1} (1 - \alpha_j)$$

5. Loss Computation

Compare the rendered image I_{rendered} to the ground truth image I_{GT} . Compute:

$$\mathcal{L}_1 = \sum_{u,v} \|I_{\text{rendered}}(u,v) - I_{\text{GT}}(u,v)\|_1$$

Additionally, use the D-SSIM metric:

- Structural similarity metric for image patches.
- Sensitive to blurring/texture.

The final loss is computed as:

$$\mathcal{L} = (1 - \lambda) \cdot \mathcal{L}_1 + \lambda \cdot \mathcal{L}_{\text{DSSIM}}, \quad \lambda = 0.2$$

6. Backward Pass – Gradient Computation

Compute gradients for all parameters:

- **Position:** Moves the Gaussian in 3D space.
- **Scale and Rotation:** Adjusts shape/orientation.
- **Opacity:** Controls the contribution to alpha-blending.
- **SH Coefficients:** Adjusts directional color.

Efficient memory trick:

- Store the final accumulated alpha during the forward pass.
- Reuse the sorted Gaussian lists to compute gradients back-to-front.

7. Parameter Updates

Update parameters using the Adam optimizer:

- Separate learning rates for:
 - Position (high),
 - SH coefficients (medium),
 - Opacity (low).
- Apply gradient steps:

$$\theta \leftarrow \theta - \eta \cdot \nabla_{\theta} \mathcal{L}$$

Clamp or reparameterize:

- Opacity via sigmoid.
- Scale via exponential.

- Normalize quaternions to maintain valid rotations.

8. Adaptive Density Control

Every N iterations (e.g., 100):

- **Clone** Gaussians if they have small scale, large view-space gradient magnitude, or exist in under-reconstructed regions.
- **Split** Gaussians if they have large scale, high error, or span multiple surfaces.
- **Prune** Gaussians if opacity is below threshold, footprint is massive, or they do not contribute to the error.

This dynamically sculpts the cloud of Gaussians.

9. SH Expansion

Every 1,000 iterations:

- Add one band of SH.
- Gradually increase from one coefficient (band 0) to 16 coefficients (band 3).

10. Resolution Upscaling

Training begins at low resolution:

- Upscale from $1/4$ size $1/2$ full resolution.

11. Convergence

After approximately 5,000–30,000 iterations:

- The loss plateaus.
- Gaussians settle into a detailed, optimized distribution.

The final scene contains 100k–500k Gaussians, each with optimized shape, pose, and SH color.

12. Output

Upon completion, save the following:

- **Scene file:**
 - Gaussian data (μ, Σ, α, SH) typically occupies a few hundred MB.
- **Renderer:**
 - GPU tile rasterizer capable of rendering at 30–150 FPS at 1080p.

Conclusion

You have successfully trained a 3D scene from 2D images using Gaussians, resulting in a compact 3D representation that supports real-time, view-dependent rendering without the need for meshes, voxels, or heavy MLPs.

Table 2.1: TL;DR Pipeline Overview

Stage	Description
1. Input	2D images + SfM-calculated camera poses and point cloud
2. Init	Turn each 3D point into a Gaussian with position, shape, color
3. Projection	Project Gaussians into camera space, splat to image plane
4. Rendering	Rasterize using tile-based GPU renderer
5. Loss	Compare rendered image to ground truth using L1 + D-SSIM
6. Backprop	Compute gradients, update Gaussian parameters
7. Density Control	Clone/split/prune Gaussians adaptively
8. Repeat	Iterate thousands of times
9. Output	A real-time, high-quality Gaussian splat scene

Chapter 3

Related Works

The domain of 3D Gaussian Splatting (3DGS) in medical imaging represents a confluence of advances in volumetric rendering, real-time visualization, and medical data compression. This section presents a synthesized and critically analyzed narrative of key works in this emerging field, focusing particularly on three pivotal contributions: Cinematic Gaussian Splatting (Cinematic GS), Gaussian Splatting for CT (GaSpCT), and Multi-Layer Gaussian Splatting. These studies, while building on a shared foundational technique, diverge significantly in their objectives, methodologies, and addressed challenges, offering a rich landscape for comparison and highlighting critical gaps for future exploration.

3.1 Evolution of 3DGS in Medical Imaging

The genesis of 3D Gaussian Splatting lies in the broader field of neural rendering and real-time graphics, where the need to balance visual fidelity with computational efficiency has long been a central challenge. Early volumetric rendering techniques in medical imaging, such as ray casting and direct volume rendering, provided high-quality visualizations but at the cost of significant computational expense. Traditional volume rendering pipelines often struggled with the massive size of medical datasets, particularly high-resolution CT and MRI scans, limiting their real-time applicability in clinical and educational settings.

Cinematic Gaussian Splatting, as proposed by Niedermayr et al., marks a critical leap by compressing gigabyte-scale volumetric datasets into lightweight, GPU-optimized Gaussian representations. Simultaneously, GaSpCT reimagines the problem of CT reconstruction itself, adapting Gaussian Splatting not for visualization but for data

recovery from sparse X-ray projections. Multi-Layer GS, in turn, extends the cinematic approach by introducing a layered structure that permits partial interactivity in VR exploration of anatomy.

Together, these works redefine the possibilities of real-time, high-fidelity medical imaging, each approaching from a distinct perspective: visualization, reconstruction, and interaction.

3.2 Methodological Innovations and Divergences

Although grounded in a shared philosophical foundation—the representation of volumetric medical data through ensembles of anisotropic Gaussians—the methodologies adopted by Cinematic Gaussian Splatting, GaSpCT, and Multi-Layer Gaussian Splatting diverge markedly based on their target applications, optimization goals, and underlying assumptions about the data.

At a high level, each approach demonstrates a distinct answer to the question: How can the flexibility of Gaussian primitives be harnessed to overcome the challenges inherent in medical imaging? By examining their pipelines, loss formulations, architectural choices, and optimization strategies in detail, we can appreciate the rich methodological landscape emerging around 3DGS in medical applications.

3.2.1 Cinematic Gaussian Splatting: Optimizing for Static Photorealism

Cinematic Gaussian Splatting is designed foremost for static, high-fidelity anatomical visualization. Its pipeline is meticulous and computation-heavy, prioritizing visual realism and frame rate performance over dynamic interaction or data efficiency.

The process begins with offline high-quality rendering, typically through path tracing or ray-marching techniques, to generate reference views of the target anatomy under carefully controlled lighting conditions. Recognizing that comprehensive anatomical coverage cannot be achieved through random sampling alone, strategic camera pose selection becomes critical: multiple viewpoints are algorithmically determined to capture all surfaces and cavities, ensuring no anatomical structure is occluded or lost during optimization.

Subsequently, the volumetric field is represented through millions of anisotropic Gaussians, each parameterized by position, orientation, color, opacity, and scale. Optimization then proceeds iteratively, aligning the projections of these Gaussians with

the reference images via photometric loss functions. To support efficient real-time rendering, mip-splatting techniques are introduced, adapting the density and resolution of Gaussians dynamically based on camera distance and scene complexity. This level-of-detail control ensures that computational resources are directed only where needed, allowing cinematic models to run smoothly even on limited hardware like standalone VR headsets.

However, a major limitation is embedded in the methodology: the final model encodes static lighting, static transfer functions, and static anatomical composition. Any change in viewing conditions not captured during the offline phase necessitates a complete re-optimization, rendering the approach less suited for exploratory or diagnostic applications.

3.2.2 GaSpCT: Data-Efficient Reconstruction from Sparse Projections

In stark contrast to Cinematic GS’s focus on visualization from full volumes, GaSpCT reorients Gaussian Splatting toward the problem of data reconstruction—specifically, recovering volumetric representations from limited-angle 2D X-ray projections.

GaSpCT’s pipeline starts not with a complete 3D scan but with a sparse, incomplete set of X-ray images, often acquired under severe angular limitations (e.g., due to patient motion restrictions, radiation dose constraints, or mechanical limits of portable scanners). Unlike Cinematic GS, where millions of Gaussians are initialized around known volumetric surfaces, GaSpCT must hallucinate structure from minimal input.

To address this, the method introduces uniform ellipsoidal initialization of Gaussians throughout the imaging volume. Instead of placing Gaussians on known surfaces, GaSpCT initializes a dense grid of small, axis-aligned ellipsoids that uniformly tile the entire 3D domain. The optimization objective is then twofold:

- Minimize photometric error between the rendered projections of the Gaussians and the acquired X-ray images.
- Encourage anatomical sparsity and continuity via regularization terms.

The innovation lies in its loss design:

- The `beta sparsity loss` promotes selective opacity, penalizing unnecessary Gaussian contributions and driving the model toward a compact, anatomically plausible structure.

- Total variation loss enforces smoothness across neighboring Gaussians, discouraging noisy or fragmented reconstructions.

Furthermore, the rendering model itself is modified: Gaussians are rendered under a pinhole camera approximation rather than complex volume scattering models, simplifying the optimization and speeding up training.

Through these innovations, GaSpCT achieves the remarkable ability to reconstruct detailed volumetric approximations with only a fraction of the imaging data traditionally required. Yet, the trade-off is evident: fidelity and accuracy are limited by the assumptions embedded in the losses and the relatively simple initializations, and the method’s success is highly dependent on the quality and diversity of the sparse views.

3.2.3 Multi-Layer Gaussian Splatting: Introducing Modularity and Interactive Control

Building on the successes of Cinematic GS in visualization but recognizing its limitations in dynamic interaction, Multi-Layer Gaussian Splatting proposes a radical extension: the decomposition of anatomy into hierarchical, semantically meaningful layers.

The pipeline of Multi-Layer GS begins similarly with a high-resolution volumetric input. However, instead of a monolithic optimization over a single set of Gaussians, the volume is first segmented—either manually or algorithmically—into distinct anatomical structures or functional groups (e.g., skeletal system, vascular network, organ systems). Each segment is then assigned a dedicated transfer function defining its optical properties.

Optimization proceeds independently for each layer, enabling fine-grained control over opacity, color, and density without interference between structures. Importantly, this layered organization allows for real-time toggling, clipping, or recombination of anatomical groups during VR exploration. Users can selectively hide muscle tissue to reveal the skeletal system beneath, or isolate specific organ groups for detailed study.

Nevertheless, layering introduces a significant challenge: memory overhead. Each anatomical layer increases the number of Gaussians in the scene, quickly leading to unsustainable growth in model size and rendering cost. To mitigate this, Multi-Layer GS incorporates two key techniques:

- **Inactive Gaussian Pruning:** Gaussians that contribute negligibly to rendered views or reside entirely inside occluded structures are identified and removed

post-optimization.

- **Gaussian Clustering:** Redundant or spatially proximate Gaussians are merged into single, composite Gaussians, reducing computational load while preserving overall visual fidelity.

This modular approach fundamentally shifts the design space of 3DGS applications, moving from static snapshots to semi-dynamic, user-driven exploration—albeit still constrained by the need for precomputed layers.

3.2.4 Synthesis of Methodological Themes

Across Cinematic GS, GaSpCT, and Multi-Layer GS, a coherent narrative of innovation emerges. Each method strategically tailors the flexible Gaussian representation to prioritize different axes of performance:

- Cinematic GS maximizes visual fidelity and compression efficiency, sacrificing adaptability.
- GaSpCT optimizes for data-efficiency and robustness to sparse input, accepting approximate reconstructions.
- Multi-Layer GS trades off absolute visual fidelity to regain modularity and interaction.

Collectively, they demonstrate the remarkable adaptability of 3D Gaussian Splatting—but also highlight its current boundaries: struggles with dynamic lighting, difficulty in scaling to arbitrary interactivity, and challenges in balancing memory usage against realism.

By deeply understanding these divergences, we can better identify how to build the next generation of Gaussian-based models that unify fidelity, efficiency, and flexibility into a cohesive, deployable technology for medical imaging.

3.3 Comparative Analysis: Applications, Strengths, and Weaknesses

A closer examination of the three approaches reveals nuanced similarities and contrasts:

Despite sharing a common technological core, the three methods serve different facets of medical imaging needs: high-fidelity visualization (Cinematic GS), data-efficient

Table 3.1: Comparative analysis of the key aspects of Cinematic GS, GaSpCT, and Multi-Layer GS

Aspect	Cinematic GS	GaSpCT	Multi-Layer GS
Input Data	High-res volumetric CT/MRI	Sparse 2D X-ray projections	High-res volumetric CT
Goal	Photorealistic anatomy visualization	Sparse-view CT reconstruction	Interactive VR anatomy exploration
Key Techniques	Mip-splatting, camera pose optimization	Beta sparsity loss, uniform Gaussian initialization	Layered training, Gaussian clustering and pruning
Strengths	High fidelity, low memory footprint	Reduced scan burden, efficient training	Moderate interactivity, layered anatomy visualization
Weaknesses	Static visualization, fixed lighting	Sensitivity to loss design, brain-specific focus	Static layers, performance drop with deep zoom

image synthesis (GaSpCT), and interactive exploration (Multi-Layer GS).

However, all three methods grapple with inherent limitations of current 3DGS techniques: static representations, fidelity versus compression trade-offs, heavy preprocessing, and challenges in generalization across modalities and patient populations.

3.4 Identified Research Gaps and Opportunities

While the emergence of Gaussian Splatting techniques—namely Cinematic Gaussian Splatting, GaSpCT, and Multi-Layer Gaussian Splatting—has significantly advanced the frontier of medical image visualization and reconstruction, **critical** limitations persist across these methodologies. A careful synthesis of the existing literature reveals several pivotal gaps that hinder the broader clinical and research adoption of 3DGS-based systems. In particular, our work focuses on addressing the fundamental constraint imposed by orthographic projection assumptions, and reimagining Gaussian Splatting for dynamic, interactive, and clinically robust settings.

3.4.1 Static Volumetric Representations and Lack of Interactivity

A central and recurring limitation across all existing Gaussian Splatting applications in medical imaging is the reliance on static volumetric representations. Once optimized, the Gaussian field becomes fixed: lighting conditions, material transfer functions, slicing planes, and structural visibility are all baked into the model during offline training. Consequently, any user-driven interaction that alters the rendering conditions—such as changing lighting angles, highlighting different tissue densities, or interactively clipping anatomical structures—requires either substantial approximation or complete re-optimization of the Gaussian parameters.

This immutability fundamentally restricts the usability of 3DGS models in clinical and educational environments, where dynamic, real-time exploration of anatomy is often essential. For example, a radiologist examining tumor margins may wish to dynamically adjust contrast settings; an anatomy student may need to clip through

layers to better understand spatial relationships. Existing models’ inability to adapt to such exploratory interactions in real-time constitutes a major barrier to practical deployment.

3.4.2 Absence of Dynamic (4D) and Temporal Support

Beyond static interactivity, there is also a profound absence of support for time-varying phenomena in current 3DGS frameworks. None of the surveyed methods account for dynamic processes such as cardiac motion, respiratory cycles, contrast-agent flow, or longitudinal disease progression. These temporal aspects are central to a wide range of clinical applications—including functional imaging, perfusion studies, and intraoperative navigation—and their omission severely limits the relevance of current splatting techniques to real-world medical workflows.

Integrating 4D information into the 3DGS paradigm presents considerable technical challenges, from data alignment and motion modeling to memory and rendering efficiency. Nonetheless, addressing this gap is critical for the next evolution of Gaussian-based medical visualization.

3.4.3 Data Domain Limitations and Lack of Multimodal Support

Another key constraint is the narrow data domain explored by existing works. Current methodologies are predominantly validated on relatively clean, high-resolution CT datasets, often derived from controlled research collections. Little to no exploration has been conducted into other clinically significant modalities such as MRI, PET, or ultrasound, each of which brings unique challenges like lower signal-to-noise ratios, modality-specific artifacts, and vastly different tissue contrast profiles.

Moreover, real-world clinical data often exhibits issues such as motion blur, incomplete coverage, metal artifacts, and varying image quality—all factors minimally addressed in current Gaussian Splatting literature. Without demonstrating robustness across diverse, noisy, and artifact-laden datasets, existing methods remain largely academic, limiting their immediate clinical translatability.

3.4.4 Performance vs. Fidelity Trade-offs

While 3DGS models excel in achieving real-time rendering through aggressive data compression, this often comes at the expense of anatomical fidelity—especially in

complex, thin, or semi-transparent structures such as vasculature, nerve bundles, or micro-calcifications. For many diagnostic applications, even minor losses of detail can have serious clinical implications.

Current Gaussian pruning, clustering, and level-of-detail management strategies prioritize rendering efficiency but do not yet offer nuanced mechanisms to preserve critical diagnostic features. Striking a better balance between performance and fidelity remains an open research problem, particularly for applications where clinical trust and precision are paramount.

3.4.5 Heavy and Inflexible Offline Preprocessing

The generation of current 3DGS models involves heavy offline preprocessing pipelines, including computationally expensive path-traced reference generation, millions of optimization iterations, and hand-crafted pose sampling strategies. This process can take many hours to days, depending on dataset complexity and hardware resources.

In clinical contexts—where speed, scalability, and cost-efficiency are vital—such prolonged preprocessing times are impractical. Reducing this preprocessing bottleneck, or enabling incremental updates without full retraining, would markedly improve the feasibility of Gaussian Splatting technologies in real-world workflows.

3.5 The Critical Limitation: Orthographic Projection Assumptions

One fundamental technical limitation underlying many of these challenges—particularly in GaSpCT and related methods—is the implicit reliance on orthographic or simplified projection models during both training and inference.

In GaSpCT, for instance, the use of pinhole approximations or simplified orthographic projections during Gaussian optimization provides computational efficiency and mathematical tractability. However, this assumption fails to capture the true perspective distortions inherent in medical imaging modalities such as cone-beam CT (CBCT), fluoroscopy, or X-ray angiography.

3.5.1 Orthographic approximations:

- Oversimplify the projection geometry, leading to inaccuracies in the reconstructed volumetric fields—especially for wide-angle scans or off-center anatomical re-

gions.

- Limit the generalizability of models across different scanner configurations and patient setups.
- Hinder multi-modal fusion, where different modalities (e.g., PET-CT) require precise geometric correspondence across complex perspective transformations.

In essence, orthographic projection assumptions act as a bottleneck, constraining the fidelity, flexibility, and clinical applicability of Gaussian Splatting-based systems.

3.6 Opportunities for Advancement

Our work directly addresses this pivotal shortcoming by introducing perspective-aware Gaussian Splatting frameworks. By modeling the full cone-beam geometry during Gaussian rendering and optimization, we aim to:

- Improve volumetric reconstruction accuracy across varying scanner types and imaging geometries.
- Enable cross-modal compatibility, ensuring geometric consistency across diverse imaging modalities.
- Enhance dynamic interaction, by supporting real-time transformations that respect perspective effects.

Moreover, we are pioneering approaches that integrate dynamic lighting and slicing at inference time without retraining, through modular scene parameterization and differentiable compositing strategies.

By systematically dismantling the orthographic assumption and building Gaussian Splatting systems that natively handle full projection physics and dynamic control, we seek to bridge the critical gap between academic innovation and clinical utility.

3.7 Synthesizing the Trajectory: A Cohesive Narrative

The evolution of Gaussian Splatting in medical imaging reflects a broader technological narrative: the transition from static, heavy volumetric datasets to agile, high-fidelity, real-time representations capable of supporting next-generation clinical and educational applications.

Cinematic GS pioneers the foundation, demonstrating that gigabyte-scale medical volumes can be condensed into lightweight models while preserving photorealism. GaSpCT reorients this approach toward data efficiency, leveraging the flexibility of Gaussian fields to reconstruct images from sparse inputs—a step toward reducing radiation burden. Multi-Layer GS pushes further, seeking to restore a measure of interactivity into otherwise static models by layering anatomical structures.

Despite their progress, all current approaches remain constrained by static designs, domain specificity, and precomputation bottlenecks. Future research must therefore seek to transcend these barriers, evolving Gaussian Splatting into a dynamic, generalizable, and clinically robust paradigm.

3.8 Conclusion: Toward a Dynamic Future for 3DGS in Medical Imaging

In summary, while Cinematic GS, GaSpCT, and Multi-Layer GS each significantly advance the capabilities of 3D Gaussian Splatting in medical imaging, they also expose fundamental limitations that circumscribe the current state of the field. Static models, limited modality adaptation, fidelity-performance compromises, and heavy preprocessing collectively represent the critical frontiers yet to be conquered.

The next generation of 3DGS research must therefore focus on enabling real-time dynamic interactions, multi-modal adaptability, efficient compression without sacrificing detail, and scalability to clinical realities. Furthermore, innovations such as dynamic slicing, interactive relighting, 4D Gaussian splatting, and generative augmentation will be pivotal in unlocking the full potential of Gaussian Splatting for medical imaging.

By building on the strong foundations laid by these early works, the research community stands poised to redefine the landscape of medical visualization and reconstruction for the decade to come.

Chapter 4

Proposed Methodology

The Proposed Methodology chapter is the core of your thesis, where the approach to solving the research problem is presented in a detailed and structured manner. This chapter aims to provide a clear, step-by-step explanation of how you plan to address the problem outlined in the introduction. Each component of the methodology should be explained in a way that allows readers, even those less familiar with the specific technical details of the project, to understand both the structure and the rationale behind your choices. Clarity is essential, and visual aids should be used to enhance understanding where necessary.

4.1 Overview of the Methodology

This project focuses on adapting and validating the 3D Gaussian Splatting (3DGS) framework for the specific challenges posed by medical imaging, particularly CT scan slice datasets. Unlike conventional photographic datasets typically used in Gaussian Splatting, medical datasets feature orthographic projections and lack distinctive overlapping features between views, making direct application of standard 3DGS infeasible.

To systematically address these challenges, we structured the methodology into several interconnected phases. These include dataset acquisition and preprocessing, establishing a functional Gaussian Splatting pipeline adapted to our environment, generating synthetic camera models suited to orthographic slice data, modifying the splatting process to account for these projections, and finally exploring alternative strategies to overcome inherent limitations.

Each phase builds upon the previous one, forming a comprehensive approach that

addresses both the theoretical and practical barriers in adapting 3DGS for medical contexts.

4.2 Chronological Breakdown of Components

4.2.1 Dataset Acquisition and Preprocessing

At the outset, a major challenge was sourcing an appropriate dataset. Our primary need was for a stack of medical images representing sequential anatomical slices, suitable for volumetric reconstruction. Although several datasets were evaluated, most were tailored either for diagnostic purposes or were heavily post-processed, limiting their utility for our goals. X-ray datasets were initially considered but soon discarded, as their 2D projections lacked the necessary volumetric depth.

Ultimately, we selected a Hip-CT dataset comprising approximately 1800 slices. However, the dataset was provided in JPEG2000 (.jp2) format, which was incompatible with our planned pipeline. A two-step conversion process was therefore implemented: the .jp2 images were first converted to TIFF (.tif) format to maintain quality, and then to 8-bit PNG images to reduce file size and ensure compatibility with the Gaussian Splatting input pipeline.

Throughout preprocessing, careful attention was paid to preserving slice order and pixel integrity to maintain anatomical correctness across the volume.

4.2.2 Establishing a Functional Gaussian Splatting Pipeline

Following dataset preparation, the next step was to set up a functional environment capable of running a 3D Gaussian Splatting pipeline. We selected the codebase from Gaussian Splatting for Cinematic Anatomy as our starting point. However, significant difficulties arose during environment setup. Although the authors provided an environment.yml file, it contained multiple inconsistencies, particularly relating to CUDA, PyTorch, and TorchVision versions.

Through a painstaking manual process involving dependency resolution, CUDA downgrades, package version pinning, and Visual Studio compatibility adjustments, a functional environment was eventually established. Additionally, submodules used by the codebase required their own specific configurations, adding further complexity.

To validate the setup, we initially ran experiments using an M60 tank dataset, for which a full Structure-from-Motion (SfM) solution already existed. Successful exe-

cution of the splatting pipeline on this dataset confirmed that the base functionality was intact, including camera pose handling, point cloud optimization, and splat generation.

4.2.3 Addressing Structure-from-Motion Challenges in CT Slices

Having established that the splatting pipeline worked for conventional datasets, we turned our attention to the medical CT slices. Unlike photographs, CT slices are essentially orthogonal cross-sections with no overlap or shared features between adjacent images. When we attempted to run classical SfM algorithms such as COLMAP on the CT slices, the system was able to find common features between only two images out of 1800 — a clear indication that the standard feature matching approach was untenable.

Recognizing this, we developed a method for synthetic camera pose generation. Given that the CT slices represent sequential slices through the anatomy, we synthesized extrinsic camera parameters by placing virtual cameras along a single axis at regular intervals, corresponding to assumed slice thickness. For the intrinsic parameters, we initially assumed a standard perspective camera model, defining a plausible field of view based on the slice dimensions.

However, early experiments revealed that this assumption was incorrect, as CT scanners effectively operate under an orthographic projection model, not a perspective one. This realization led to a significant revision of our camera model, detailed in the next section.

4.2.4 Camera Model Adaptation: Perspective to Orthographic Projection

Our first attempt at synthetic camera modeling was based on the premise of perspective projection. Virtual cameras were placed at incremental positions along the Z-axis, each directed orthogonally to the imaging plane, with a field of view consistent with standard camera models. When these parameters were used for splatting, the resulting point cloud resembled a conical burst: Gaussians were densely clustered near the apex and sparsely distributed elsewhere, bearing no resemblance to the underlying anatomical structure.

On further analysis, we concluded that CT scanners generate images with parallel projection rays, implying an orthographic rather than perspective geometry. Conse-

quently, we rewrote our camera generation scripts to model an orthographic projection, where scale remains constant with depth and rays do not converge.

In the orthographic setup, each camera was defined with:

- A simple linear translation along the slicing axis.
- Constant scale factors across all views.
- Parallel projection without foreshortening.

Although this correction produced better-structured splat distributions, substantial challenges remained, particularly regarding point density and anatomical fidelity, due to the random initialization of the splatting pipeline.

4.2.5 Feeding Medical Data into the Pipeline

With orthographic camera models in place, the medical dataset was fed into the Gaussian Splatting pipeline. Several modifications to the original code were necessary:

- Removal of assumptions requiring textured background images.
- Adjustment of point cloud initialization parameters to better match uniform volumetric distributions.
- Extensive increases to the number of training iterations (up to 200,000) to compensate for the lack of natural features and to allow the optimizer to converge on a meaningful distribution.

Despite these adaptations, the optimization process struggled to correctly infer the internal structures of the volume. The point cloud often collapsed at the center or formed random dense blobs at the edges, reflecting the limitations of applying a perspective-trained optimization strategy to orthographic data without intrinsic depth variation cues.

4.2.6 Exploring Alternative Strategies

Given the limitations observed, we initiated exploration of complementary approaches to improve the representation of medical datasets.

Firstly, Napier was used to generate voxelized density fields from the CT slice stacks. These voxel representations inherently carry 3D spatial relationships and could potentially serve as intermediate inputs for splatting, bypassing the need for feature match-

ing altogether. Future work includes generating synthetic views from voxel maps, which can then feed into a modified splatting pipeline.

Secondly, a deeper exploration of DICOM datasets is underway. Unlike basic PNG or JPEG2000 images, DICOM files often encode detailed metadata, including pixel spacing, slice thickness, and scanner geometries. Extracting this information could enable the generation of more accurate camera intrinsics and extrinsics without relying on manual assumptions.

Lastly, work is ongoing toward incorporating GPU-accelerated rasterization techniques to speed up point cloud visualization and training iterations, making the process more interactive and scalable for larger datasets.

4.2.7 Full Breakdown of Our Adapted Pipeline

The adapted pipeline developed during this project comprises several carefully engineered stages, each designed to bridge specific gaps between conventional 3D Gaussian Splatting workflows and the unique requirements of medical imagery.

The full sequence is as follows:

1. Data Collection and Preprocessing:

- Acquisition of raw CT slices (Hip-CT dataset) originally in JPEG2000 (.jp2) format.
- Conversion to TIFF format to maintain quality, followed by conversion to 8-bit PNG for compatibility.
- Manual verification of slice order and pixel integrity to preserve anatomical consistency.

2. Environment and Codebase Setup:

- Cloning and setting up the Cinematic Gaussian Splatting codebase.
- Extensive manual dependency resolution: CUDA downgrades, PyTorch and TorchVision version alignment, Visual Studio compatibility fixes, and submodule configuration.
- Validation of functionality using a conventional M60 Tank dataset with precomputed SfM data.

3. Initial Attempt at Structure-from-Motion (SfM) on Medical Data:

- Direct application of COLMAP on CT slices.
- Observation: Traditional SfM failed due to absence of overlapping features across slices.

4. Synthetic Camera Pose Generation:

- Assumption of initial perspective camera model; placement of virtual cameras along the slicing axis.
- Manual generation of camera intrinsics (focal length, principal point) and extrinsics (translation along the Z-axis).
- Identification of projection model mismatch after initial splatting failures.

5. Orthographic Camera Model Adaptation:

- Correction of the camera model to orthographic projection.
- Removal of perspective distortion effects, ensuring parallel projection rays.
- Re-generation of camera intrinsics and extrinsics accordingly.

6. Feeding Medical Data into the Splatting Pipeline:

- Modification of splatting code to remove hard dependencies on background images and perspective assumptions.
- Adaptation of point cloud initialization parameters for better volumetric uniformity.
- Execution of long training runs (up to 200,000 iterations) to allow optimization over challenging orthographic data.

7. Evaluation and Visualization:

- Analysis of generated Gaussian point clouds.
- Recognition of limitations: lack of central structure, collapse of splats at edges due to sparsity cues.
- Use of third-party web-based viewers for qualitative assessment.

8. Alternative Strategy Exploration:

- Deployment of Napier voxelization tools to generate intermediate 3D density fields.

- Initiation of studies into DICOM metadata extraction for more reliable camera modeling.
- Preliminary steps toward GPU-based rasterization to accelerate iterative visualization.

Each of these stages is interconnected, with insights from failures and successes at each step feeding back into iterative improvements across the pipeline. The careful modular design of this adapted pipeline ensures that individual components can be independently refined or replaced as better solutions are developed.

4.3 Summary of the Methodology

This chapter has detailed the comprehensive and adaptive methodology used to tackle the complex problem of applying 3D Gaussian Splatting to medical imaging datasets. Beginning with careful dataset curation and preprocessing, progressing through establishing a working splatting pipeline, adapting the camera models for orthographic projection, and ultimately exploring alternative representation strategies, each step has been carefully considered to address both theoretical and practical challenges.

This systematic methodology forms the foundation for the experiments and evaluations that follow.

Chapter 5

Results and Discussion

The Results and Discussion chapter is where you present the key findings of your research and analyze their significance in relation to your research objectives. Whether combined or separated into two distinct sections, this chapter must go beyond simply displaying data, providing a detailed interpretation that connects the results with theoretical frameworks and empirical justifications. The choice to combine or separate the sections depends on the complexity of your findings, but in either format, this chapter should offer a compelling narrative that illuminates the implications of your work.

5.1 Datasets and Experimental Setup

In this subsection, we will outline the datasets used for the experiments and describe the experimental setup, including the challenges faced in preprocessing, environment setup, and model training. The choice of dataset and the setup for the experiments were critical in testing the viability of Gaussian splatting for medical image processing.

5.1.1 Datasets Used

Medical Dataset (CT Scan Slices)

Dataset Description: Our primary medical dataset consisted of 1800 slices from a CT scan focused on the partial side of the brain. This dataset was chosen because of its potential to provide detailed cross-sectional images of human anatomy, which are often used in medical visualizations for 3D reconstruction.

Preprocessing Challenges:

- The original dataset was in JPEG2000 (.jp2) format, which was not directly compatible with Gaussian splatting. We had to convert the images into an intermediary TIFF (.tif) format before finally converting them to PNG format, an 8-bit representation, to fit the requirements of the pipeline.
- **Data Size:** The dataset consists of 1800 2D slices, each capturing a small section of the brain. These slices need to be processed in a way that allows them to be treated as a 3D volumetric representation, which posed unique challenges for alignment and 3D reconstruction using Gaussian splatting.

M60 Tank Dataset (for Proof of Concept)

Dataset Description: To validate the pipeline’s basic functionality, we utilized the M60 tank dataset. This dataset consists of 300 images of a tank from different angles. Unlike the medical dataset, these images were pre-aligned, with computed camera extrinsics and intrinsics provided.

SfM Calibration: Since the dataset came with precomputed camera parameters, it was particularly useful for testing the pipeline’s ability to generate 3D models from 2D images without having to manually solve for camera extrinsics. This allowed us to focus on testing the Gaussian splatting algorithm itself without the need to worry about camera calibration.

Challenges with Dataset Compatibility

Format Issues: One of the major challenges we faced was the incompatibility of standard medical imaging formats like JPEG2000 with the Gaussian splatting pipeline. A significant amount of time was spent converting the images into a format suitable for processing by the pipeline.

Missing Slice-to-Slice Correspondences: A major obstacle was the nature of the CT slices themselves. Since the images were sequential 2D slices taken from various angles of the brain, they lacked shared features that would allow for traditional Structure from Motion (SfM) methods to work. As a result, camera parameters had to be manually generated, which introduced a degree of error into the pipeline.

5.1.2 Experimental Setup

Preprocessing and Environment Setup

Initial Challenges: Setting up the environment for Gaussian splatting involved many challenges, starting with dependency management. The official codebase from Gaussian Splatting for Cinematic Anatomy was retrieved, but it lacked a working environment setup. The environment.yml file provided by the authors contained several outdated or incompatible package versions, which resulted in numerous dependency issues.

- We had to roll back certain versions of CUDA, PyTorch, and TorchVision to make the pipeline functional.
- Additionally, we discovered that Visual Studio 2022 was incompatible with the pipeline, requiring us to use an older version to get the environment working.

Final Setup: After many iterations of trial and error, we successfully resolved the dependency conflicts and had a working environment, enabling us to move forward with the preprocessing and training stages.

Camera Calibration and SfM

SfM Issues with CT Scans: One of the key hurdles in adapting Gaussian splatting to medical datasets was solving for the camera parameters. The traditional SfM approach relies on finding common features (edges, textures) between multiple images to align them in 3D space. However, with CT slices, there were no shared features between adjacent slices, rendering standard SfM methods ineffective.

Fake Camera Extrinsics: To bypass this limitation, we generated fake camera extrinsics by assuming a certain slice thickness. This allowed us to approximate the relative positions of the camera for each slice. While this method was not accurate, it provided enough structure to test the feasibility of applying Gaussian splatting to medical data.

Perspective vs. Orthographic Projection: Another complication arose when we assumed that the CT slices followed a perspective projection model, which is typically used in photography. However, CT scans actually use an orthographic projection, which significantly altered the behavior of the Gaussian splatting algorithm. This realization led us to modify our camera extrinsics script to account for orthographic projections, which was a more suitable model for the data.

Model Training and Fine-tuning

Training Setup: The Gaussian splatting model was trained using GPU-based acceleration, given the computational intensity of the process. The initial model training involved adjusting key hyperparameters such as the number of iterations and the density threshold.

- We started with a base set of training parameters and gradually adjusted them through trial and error. For the M60 tank dataset, training took approximately 22.33 minutes with 100,000 iterations, which produced a rudimentary 3D point cloud that later became a Gaussian splat.
- When we transitioned to the medical dataset, we initially used the same training parameters, but the results were far less promising due to the errors introduced by fake camera data and the incompatibility of the orthographic projection with Gaussian splatting.

5.2 Presenting the Results

In this section, we present and analyze the results obtained from applying the Gaussian splatting pipeline to both the M60 tank dataset and the CT scan slices dataset. The results offer insights into the effectiveness of Gaussian splatting for 3D medical visualization and highlight areas that need improvement.

5.2.1 M60 Tank Dataset: Proof of Concept

Initial 3D Reconstruction:

When applying Gaussian splatting to the M60 tank dataset, we observed a significant improvement in the output compared to earlier tests with medical slices. The process started by generating a random point cloud, which eventually converged into a more structured 3D representation after training.

Random Point Cloud: After running the model for the initial few iterations, the output was a sparse, unstructured point cloud with no clear form. However, with continued iterations, the model began to refine the cloud and generate a more cohesive 3D structure.

Gaussian Splat Representation: After training, the final output consisted of a series of Gaussian splats that were loosely distributed around the object. Although the

visualization was not perfect, it demonstrated that Gaussian splatting could generate a rudimentary 3D representation of a complex object like a tank.

Visualization:

The results were displayed using web-based third-party tools that allowed for basic visualization of the generated Gaussian splats. These tools allowed us to explore the 3D structure in a simple, but effective way, helping us to evaluate the quality of the generated point clouds.

5.2.2 CT Scan Dataset: Applying Gaussian Splatting to Medical Data

Initial Results:

When we attempted to apply the Gaussian splatting pipeline to the CT scan dataset, the initial results were far less promising. Due to the incorrect assumption of perspective projection, the output consisted of a chaotic and random 3D cloud of Gaussians with no recognizable structure.

Randomness and Chaos: After several iterations, the output remained largely random, indicating that the algorithm was not able to effectively interpret the 2D CT slices. This randomness persisted despite adjustments to the training parameters, which suggested that the camera extrinsics and projection model were not correctly configured for medical images.

Revised Camera Model and Results:

After recognizing the issue with the perspective projection assumption, we modified the camera extrinsics to account for orthographic projection. This adjustment produced more structured results, although still far from perfect.

Shape of Results: The Gaussian splats began to form a rough, 3D box-like structure, though the central Gaussians were largely collapsed, leaving only the edges densely populated. This result suggested that the algorithm was optimizing for the outermost structures rather than capturing the full volume of the data.

Further Optimizations: By increasing the number of iterations and lowering the density threshold, we were able to reduce the randomness and create more densified blobs. However, the output still lacked the fine details needed to accurately represent

the medical dataset. This pointed to the fact that the optimization process still needed refinement, especially in terms of handling the complex anatomy of the brain.

Visualizing the Results:

Despite the technical challenges, the output was visualized using available tools. Although the results were blurry and lacked detail, they provided a preliminary confirmation that Gaussian splatting could indeed be applied to medical imaging data, though further adjustments were required.

5.2.3 Key Takeaways

- **Success in Basic Functionality:** The proof of concept with the M60 tank dataset confirmed that Gaussian splatting could work for generating 3D models from multiple images, even if the initial results were rough.
- **Challenges with Medical Datasets:** The major challenges arose from the unique characteristics of medical imaging data, particularly the lack of feature overlap between adjacent CT slices and the incorrect camera projection model. These issues made it difficult to achieve the desired results, though the revised orthographic projection did lead to some improvements.
- **Future Directions:** The results underscore the need for improved camera calibration techniques for medical datasets and better optimization strategies for Gaussian splatting. Moving forward, we plan to explore DICOM datasets and investigate GPU-based rasterization techniques to improve the visual fidelity of our 3D reconstructions.

5.3 Interpreting the Results

In this section, we delve into the interpretation of the results obtained from applying Gaussian splatting to both the M60 tank dataset and the CT scan slices. The results, while promising in some areas, also revealed critical insights that shaped our understanding of the feasibility of using Gaussian splatting for medical imaging.

5.3.1 M60 Tank Dataset: Insights from the Proof of Concept

Reconstruction Process:

The initial random point cloud generated from the M60 tank dataset served as a baseline for understanding how Gaussian splatting works in practice. The fact that the system could take a random distribution of points and progressively organize them into a structured 3D representation highlighted the potential of the algorithm for reconstructing complex objects.

The iterative improvement of the point cloud over several rounds of training demonstrated the algorithm’s ability to refine noisy, low-detail outputs into more recognizable 3D shapes. This aligns with Gaussian splatting’s potential to produce high-fidelity models by optimizing the placement and density of Gaussian splats based on the 3D scene’s features.

Visualization and Interpretation of Gaussian Splats:

While the resulting Gaussian splats were not perfect, the core principle of the method was demonstrated effectively. By visualizing the splats in 3D space using web-based tools, we could observe how the algorithm’s density control affected the clarity of the 3D model. The splats concentrated around the central features of the M60 tank, suggesting that Gaussian splatting excels at capturing high-density areas and more prominent object features.

However, the visualization also exposed some limitations. The final model, though recognizable, was still a bit blurry, indicating that while the method works, further refinement is needed for producing highly detailed reconstructions.

5.3.2 CT Scan Dataset: Understanding the Challenges with Medical Data

Initial Misalignment of Camera Model:

When we first applied Gaussian splatting to the CT scan slices, the randomness observed in the output could be attributed to our initial incorrect assumption of a perspective projection model. This fundamental misunderstanding of the camera model significantly skewed the results. The random cloud of Gaussians was not capturing the anatomical features as intended because the algorithm’s optimization process relied on an incorrect assumption about how the slices were captured.

Once we shifted to an orthographic projection model, the results improved, though they were still far from ideal. The 3D box-like shape generated by the model, though an improvement, was still not representative of the medical structure we aimed to reconstruct. This suggested that Gaussian splatting, as it is currently implemented, is not yet fully optimized for medical image datasets, where precision and detail are paramount.

Optimization Behavior:

The key insight from this experiment was the observation that Gaussian splatting, in its current form, tends to prioritize the outermost layers of the dataset, which results in the central Gaussians collapsing. This behavior suggests that the algorithm's optimization process is more adept at reconstructing the outer contours of an object than its internal details.

The collapse of the central Gaussians also points to a fundamental limitation in how the algorithm handles volumetric data. Medical images, such as CT scans, require an approach that can capture the full internal anatomy, not just the surface-level features. The optimization process, which focuses on minimizing energy or fitting the data, likely struggles with datasets where internal structures are not as prominent as external features.

Implications for 3D Reconstruction of Medical Data:

The results indicate that Gaussian splatting has potential but needs significant modifications before it can be used effectively with medical imaging data. The challenges observed, including poor central Gaussian representation and the difficulties with camera model calibration, suggest that further work is required to refine the pipeline and make it compatible with the unique characteristics of medical datasets.

Critical Comparison with Traditional Methods:

When compared to traditional 3D reconstruction methods used in medical imaging, such as volume rendering or mesh-based methods, Gaussian splatting performed relatively well in terms of generating an initial 3D point cloud. However, it fell short in terms of accuracy and detail. The ability to handle complex medical data such as CT scans and MRIs with Gaussian splatting is still in its infancy, and this needs to be addressed in future iterations.

5.3.3 Overall Interpretation

Strengths of Gaussian Splatting:

The M60 tank dataset demonstrated that Gaussian splatting can effectively generate 3D representations from 2D images, albeit with certain limitations. The iterative refinement process proves that the algorithm has the potential to improve over time with proper tuning of the hyperparameters.

The fact that Gaussian splatting could generate a visual 3D model from a series of 2D slices, even though the model was imperfect, shows promise for applications in other domains, such as autonomous systems, gaming, or virtual reality, where the precision of every detail is not as critical.

Weaknesses in Medical Imaging:

Gaussian splatting's major limitation with medical datasets lies in its inability to deal with complex anatomical structures and lack of shared features between adjacent slices. Moreover, the central collapse of Gaussians is problematic for medical imaging, where fine detail, especially in internal structures, is crucial.

The orthographic projection adjustment improved the results but still did not solve the problem of accurately capturing internal structures.

5.4 Challenges and Limitations

Despite the potential demonstrated in the Gaussian splatting approach, a number of challenges and limitations surfaced throughout the process. These hurdles not only slowed down the progress but also highlighted areas where the method needs refinement for practical use in medical imaging applications.

5.4.1 Challenges Encountered

Dataset Preprocessing:

Conversion Issues: One of the primary challenges faced was the conversion of medical image data into a usable format for the Gaussian splatting pipeline. The original CT scan slices were in the JPEG2000 format, which is not widely supported by most image processing libraries. Converting them to TIFF format and then to PNG added an additional layer of complexity and introduced potential for data loss or errors during the conversion process.

Volume Reconstruction: CT scans are inherently volumetric, and representing these 2D slices as a 3D volume was non-trivial. While Gaussian splatting is excellent at processing volumetric data, it struggled with aligning slices in a way that captured their true 3D structure.

Camera Calibration and SfM Challenges:

Lack of Feature Overlap: A significant limitation was the lack of feature overlap between adjacent CT slices. In traditional SfM, images are aligned based on common features (such as edges, textures, and keypoints). However, medical slices do not have these common features, making it difficult for traditional SfM algorithms to establish correspondences between images.

Fake Camera Extrinsics: Due to the absence of reliable SfM data, we had to generate fake camera extrinsics based on assumptions about slice thickness. While this allowed us to run the pipeline, it introduced errors and inaccuracies into the reconstruction process.

Projection Model Misunderstanding: Initially, we assumed that the CT scans used perspective projections, which caused major inaccuracies in the Gaussian splatting process. Switching to an orthographic projection model improved the results but did not completely solve the issue. This misinterpretation of the camera model was a key limitation that slowed down progress.

Optimization and Convergence Issues:

Randomness in Results: Early results from the CT scan dataset were plagued by randomness and a lack of meaningful structure in the output. Despite adjusting parameters and increasing the number of training iterations, the optimization process failed to converge in a way that accurately represented the anatomical structures of the brain.

Gaussian Collapse: The issue of central Gaussians collapsing was another major hurdle. This suggested that the algorithm's density optimization favors external structures, leaving the internal anatomy poorly represented.

5.4.2 Limitations of Gaussian Splatting

Inability to Handle Internal Structures:

Gaussian splatting’s optimization algorithm, while effective at capturing surface features, is not well-suited to reconstructing internal anatomical structures. This limitation was particularly evident in the CT scan dataset, where the central regions of the volume were not captured effectively.

Scalability Issues:

While Gaussian splatting works well for smaller datasets, its scalability for larger medical datasets remains uncertain. As the number of slices increases, the computational requirements grow significantly, and the method may struggle to maintain performance and accuracy.

Dependency and Compatibility Issues:

The lack of a ready-to-use environment and the frequent dependency issues highlighted the challenges of working with open-source code. The reliance on specific versions of CUDA, PyTorch, and other libraries complicated the setup and slowed down progress, as the documentation was not always up to date.

Accuracy of Camera Parameters:

The process of manually generating camera extrinsics for CT slices was prone to error, leading to inaccurate reconstructions. Gaussian splatting depends heavily on the quality of the camera data, and without reliable camera extrinsics, the results are compromised.

5.5 Implications and Significance of Findings

The results of our experiments with Gaussian splatting in the context of medical imaging present a mixed yet promising outlook for the potential application of this technique. Although the challenges we encountered in camera calibration, slice alignment, and the representation of anatomical structures are significant, they highlight areas of opportunity for future research and development in the field of 3D medical visualization.

5.5.1 Feasibility of Gaussian Splatting for Medical Imaging

Our findings suggest that Gaussian splatting could be a viable technique for 3D visualization of medical data, particularly in applications where high-level reconstructions are sufficient and fine-grained detail may not be as critical. The success observed with the M60 tank dataset, despite its inherent differences from medical imaging data, demonstrates that Gaussian splatting is capable of generating meaningful 3D representations from 2D images.

However, applying this technique to medical datasets presents unique challenges. The CT scan slices used in this study, which have their own set of idiosyncrasies like the lack of common features for image matching and the orthographic projection, did not align well with the Gaussian splatting model as initially applied. These factors make it clear that Gaussian splatting, as it currently stands, is not fully optimized for medical imaging.

With further development, specifically in the areas of camera calibration and image alignment, Gaussian splatting could evolve into a robust tool for medical 3D rendering. This could provide new opportunities for visualizing complex anatomical structures in a more dynamic and flexible manner than traditional 3D modeling methods, potentially making it more accessible and cost-effective for medical professionals.

5.5.2 Opportunities for Improvement

There are several key areas in which the pipeline can be enhanced:

- **Camera Calibration:** One of the major limitations encountered was the difficulty in generating accurate camera parameters, particularly the extrinsics for the CT slice data. Using more advanced methods for camera calibration—such as leveraging DICOM metadata or exploring other techniques for generating accurate camera models—could vastly improve the reconstruction process and provide more reliable results.
- **Dataset Selection:** Our choice of using CT slices in PNG format, while suitable for early testing, presented challenges due to format compatibility and the absence of features needed for standard Structure-from-Motion (SfM) methods. Future work should consider using more appropriate medical datasets, such as DICOM files, which are more likely to contain the necessary metadata for robust camera calibration and alignment, improving the quality of 3D reconstructions.
- **Optimization Process:** The issue of Gaussian collapse, where internal features

were not adequately represented, needs to be addressed through improvements in the optimization process. This might include developing methods to better handle internal anatomical details and prevent the loss of data in the center of the volume.

5.5.3 Contribution to Medical Visualization

This work contributes to the ongoing exploration of advanced 3D rendering techniques in the medical field. By demonstrating the potential of Gaussian splatting for medical imaging, even with the challenges encountered, this research opens up new possibilities for improving the way medical data is visualized. Specifically, it offers a pathway toward more dynamic and interactive representations of medical volumes, which could be particularly useful in medical education, surgical planning, and patient diagnosis.

The insights gathered from this work—particularly the importance of camera model accuracy, the challenges of dealing with orthographic projections, and the need for more advanced optimization techniques—will be valuable in guiding future research in 3D medical imaging. The findings also contribute to the broader body of knowledge surrounding Gaussian splatting, providing new directions for refining and expanding its applicability in complex fields like healthcare.

5.6 Decision to Split or Combine Results and Discussion

Given the iterative and technical nature of this research, combining the Results and Discussion sections seems to be the most effective approach for this thesis. This structure allows for a seamless connection between the results obtained and the challenges faced in real-time, which enhances the narrative flow and makes it easier for the reader to follow the development of the research process.

- **Direct Connection Between Results and Challenges:** Since the research involved a significant amount of trial and error, particularly in terms of code adaptation, camera calibration, and dataset conversion, it makes sense to immediately connect the results with the challenges encountered during the process. This also allows us to highlight how each challenge contributed to the overall understanding of the technique and informed the modifications made to the pipeline.

- **Fluid Narrative:** The decision to combine results and discussion ensures that readers will be able to follow a fluid and logical progression from problem identification, through experimentation, to insights and conclusions. This also helps to mitigate the feeling of fragmentation that may arise if the sections were separated.
- **Audience Engagement:** Presenting the results and discussion together makes the thesis more engaging, particularly for readers who may not be experts in Gaussian splatting or medical imaging. By providing context and interpretation side-by-side, we are able to make the research more accessible and comprehensible to a wider audience.

5.7 Conclusion of the Chapter

In conclusion, this chapter has presented a detailed examination of the experimental process, the results obtained, and the challenges encountered during our exploration of Gaussian splatting for 3D medical visualization. Despite the various hurdles, the work has yielded several important insights and opened up new avenues for future research.

- **Proof of Concept:** The successful generation of a 3D point cloud from both the M60 tank dataset and CT scan slices serves as a proof of concept for the application of Gaussian splatting to 3D medical visualization. While the generated models were not fully satisfactory, they demonstrated that the method holds promise and could be further refined with additional development.
- **Challenges and Solutions:** A series of technical challenges, particularly around camera extrinsics, the lack of common features in CT scan slices, and issues with the Gaussian optimization process, highlighted the limitations of the current pipeline. However, these challenges also offered valuable insights into the areas requiring further attention, such as camera calibration, dataset selection, and optimization techniques.
- **Future Directions:** Moving forward, the focus will shift to improving the pipeline's compatibility with medical datasets, enhancing the accuracy of camera models, and addressing the issues identified in the training and optimization process. Additionally, exploring advanced methods for optimizing Gaussian splatting for internal anatomical structures will be a priority.

In summary, while the results from this chapter are not yet fully satisfactory, they

indicate a clear path forward for using Gaussian splatting in medical imaging. The research provides a foundation for future work, with promising potential to revolutionize how medical data is visualized and analyzed in 3D environments.

Chapter 6

Conclusion

The Conclusion chapter is a critical section that wraps up the entire research process, offering both a reflection on the work completed and a forward-looking perspective. In this case, we focus on Gaussian splatting for medical imagery and the specific challenges encountered.

6.1 Restating Research Objectives and Questions

The primary objective of this thesis was to explore the feasibility of Gaussian splatting as a 3D visualization technique for medical imaging, specifically CT scan slices. This method's potential for representing complex medical data, such as brain scans, in a more dynamic and interactive way was our driving motivation.

Key research questions included:

- How can Gaussian splatting be adapted to handle medical datasets, particularly CT scan slices?
- What challenges arise in using Gaussian splatting for medical visualization, especially with respect to camera calibration and image alignment?
- Can Gaussian splatting provide better 3D reconstructions than traditional rasterization methods for medical applications?

These research questions guided our approach, focusing on testing Gaussian splatting against real medical datasets, experimenting with preprocessing, and adapting the technique to meet the unique requirements of medical imagery.

6.2 Summary of Key Findings

Our experiments provided the following key findings:

- **Feasibility of Gaussian Splatting:** The Gaussian splatting method demonstrated the potential to create 3D visualizations from medical imaging data, though initial results showed blurriness and lacked significant detail. The process did, however, validate the proof of concept.
- **Challenges with Camera Calibration:** One of the significant obstacles was the handling of camera extrinsics and the lack of standard features between CT slices. This made traditional methods like Structure-from-Motion (SfM) ineffective, requiring us to innovate new ways of faking camera data, such as assuming slice thickness for extrinsic camera parameters.
- **Orthographic Projection Complications:** Another major realization was that CT scans use orthographic projection (not perspective), which required us to rework our camera generation script to accommodate this unique property. However, traditional Gaussian splatting was still not fully compatible with this projection.
- **Data Compatibility Issues:** The Gaussian splatting pipeline, though useful for other datasets like the M60 tank dataset, struggled to adapt to the CT scan images in PNG format. This required manual data preparation, including converting image formats from JPEG2000 to PNG and making various code adjustments.

6.3 Discussion of Implications

The findings from this research have several broader implications for the use of advanced rendering techniques in the medical field:

- **Impact on Medical Imaging:** Although the results were not entirely satisfactory, they point toward the potential for Gaussian splatting to offer new ways of visualizing medical data, especially for tasks like surgical planning, education, and diagnosis.
- **Methodological Contributions:** This research contributes to the methodological understanding of Gaussian splatting in medical contexts, particularly in overcoming challenges like orthographic projections and the absence of common image features. The lessons learned in camera calibration and SfM for

medical datasets will be valuable for future studies.

- **Potential for Refinement:** With further refinement of the pipeline—especially around camera calibration, dataset format compatibility, and optimization methods—Gaussian splatting could become a more viable option for 3D medical visualization.

6.4 Contributions of the Research

The main contributions of this research are:

- **Novel Approach to Gaussian Splatting for Medical Imaging:** This work advances the understanding of how Gaussian splatting can be adapted for medical imagery, offering a novel approach to 3D medical visualization, particularly with CT scans.
- **New Insights into Camera Calibration for Medical Datasets:** By experimenting with custom scripts for generating camera extrinsics and making necessary code adaptations, we provided valuable insights into how to handle medical datasets that lack traditional features necessary for SfM.
- **Development of Preprocessing Techniques:** The work developed and tested new preprocessing techniques for preparing CT slices, including converting images to PNG format from JPEG2000 and manually creating camera extrinsics.

6.5 Acknowledging Limitations

This study is not without limitations. The main challenges we faced included:

- **Data Limitations:** The raw CT scan dataset (hip-CT) presented numerous issues, such as its compatibility with the Gaussian splatting pipeline and the lack of easily identifiable features for SfM.
- **Technical Issues with Camera Calibration:** Generating camera extrinsics for medical datasets proved much more complicated than expected, especially with orthographic projections. Our trial-and-error approach led to significant delays and results that were not fully optimized.
- **Computational Constraints:** Despite using relatively high computational resources, the iterative nature of Gaussian splatting and the density threshold adjustments resulted in limited improvement in the final visualizations.

6.6 Suggestions for Future Research

Several avenues for future research have emerged from this study:

- **Improved Camera Calibration Techniques:** Future work should focus on refining methods for generating camera extrinsics for medical datasets, possibly by integrating with existing DICOM metadata to enhance SfM accuracy.
- **Optimizing Gaussian Splatting for Orthographic Projection:** A deeper investigation into how Gaussian splatting can be made compatible with orthographic projection would be crucial for making the technique more applicable to CT scan data.
- **Alternative Datasets:** Exploring additional medical datasets, particularly DICOM images, could provide new insights into how Gaussian splatting can be better adapted for medical use cases.
- **GPU-Based Rasterization:** Given the challenges faced with traditional rasterization methods, future work could involve the development of GPU-based rasterization techniques to handle large-scale 3D data more efficiently.

6.7 Final Reflections

This research journey has been a challenging yet rewarding exploration of an emerging technique in the realm of medical visualization. We faced numerous obstacles along the way, from adapting code to dealing with incompatible datasets, but each step provided invaluable learning experiences. These challenges, although frustrating at times, allowed us to refine our approach and gain deeper insights into the complexities of medical image processing and Gaussian splatting.

6.8 Concluding Remarks

In conclusion, this study has demonstrated both the potential and limitations of Gaussian splatting for 3D medical visualization. While the results were not as expected, they have provided a solid foundation for further research in this area. By addressing the challenges identified in this work, particularly around camera calibration, data compatibility, and projection issues, future research can move closer to realizing the full potential of Gaussian splatting in medical applications.

The exploration of alternative datasets, combined with advances in camera modeling

and optimization techniques, offers a promising path forward. This research has set the stage for future work that may one day enable Gaussian splatting to play a significant role in improving medical imaging and visualization.

Chapter 7

Citations

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